

Description & Eligibility

Contestants will demonstrate practical knowledge of C# development, including core syntax, programming logic, application design, and use of objects, data access, and IDE tools within a Windows environment. This event will be graded on the Windows operating system; therefore, Unix/Linux should not be used. Any secondary or postsecondary division contestant may enter this application event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- Computer or laptop/notebook (USB-Type A or adapter required); a full-size keyboard may be used (no printer is needed) and supporting equipment (extension cords, etc.)
- Contestant must use one of these IDEs: Visual Studio 2022, Visual Studio Code, VSCode
- Only pre-written code that is hard copied is allowed

Contest Notes

- Method of evaluation: application
- AI Policy Level 1 *No AI Allowed*
- Carry-in and setup of equipment must be done solely by the contestant and within the time allotted
- Business Professionals of America assumes no responsibility for hardware/software provided by the contestant
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. Violation of this rule will result in disqualification
- At NLC, contestants will take IT Specialist - Software Development certification as part of the event. Contestants passing this certification will have 50 points added to their event score.
 - A Pearson profile must be created at <https://certiport.pearsonvue.com> before attending NLC
 - Contestants must include their BPA Contestant ID in their Pearson profile when they register
 - Certiport Parent/Legal Guardian Consent Form must be uploaded to: education@bpa.org
 - For more information on the exam, visit <https://certiport.pearsonvue.com>
 - Reference materials are not allowed for certification testing
- Length of event: no more than ten (10) minutes orientation, no more than ninety (90) minutes testing time, no more than ten (10) minutes wrap-up
 - No more than fifty (50) minutes for certification test
- Entries vary by state

Contest Competencies

1. Apply object-oriented programming principles and structured design techniques, along with efficient algorithms, to design, implement, and maintain scalable, reusable software solutions
2. Demonstrate foundational understanding of computer science concepts and proficiency in at least one modern programming language
3. Create and interpret UML diagrams to model software architecture and communicate design decisions within development teams
4. Use Entity Framework Core to design, query, and manage relational databases through object-relational mapping (ORM) techniques that support scalable application development
5. Write expressive, efficient data queries using LINQ and lambda expressions to manipulate collections, interact with databases, and simplify complex logic within .NET applications
6. Demonstrate proficiency in developing C# Windows Desktop Programming, emphasizing Windows Forms, WPF, OOP, collections, file access, debugging, and desktop UI design
7. Develop and maintain applications using updated .NET 6+ platform features, including performance enhancements, minimal APIs, modern runtime capabilities, and streamlined deployment workflows